

Course Title: Digital Programming

Course Code: EL467

Course Credit: 3-0-2-4

Prerequisite: Basic understanding of digital logic design

Course Placement: EL467 is an elective course for M.Tech. ICT (First Semester) and B.Tech. VII Sem (ICT/ICT-CS).

Course Content: The course aims to impart digital logic concept realization using hardware description language (HDL) and its practical implementation on FPGA based hardware kits. The HDL will be discussed in detail with all its abstraction levels and modeling schemes. The focus of the course will be to develop practical realizable projects using HDL and FPGA.

Sr. No.	Description
1.	Introduction to Digital Design, Concept of Hierarchical and Structured Design
	Digital meaning and essence, Basics of digital VLSI design, role of CAD tools in VLSI design process, Hierarchical and design methodologies in VLSI systems, CAD algorithms for device, circuit, gate and system levels of abstraction, Physical design process in VLSI design, Testing.
2.	Introduction to HDL, Synthesis and FPGA
	Introduction to hardware description language, Basic language elements of Verilog, Behavioral modeling, Data flow modeling, Structural modeling, Switch level modeling, Constructs and conventions in Verilog, Data types, Number system, Operators. Modeling and synthesis issues in Verilog, Familiarization with analog and mixed signal designing in Verilog. Concept of CPLD, FPGA, Familiarization with Xilinx Spartan-3E FPGA development board, design constraints using FPGA, implementing designs using FPGA.
3.	Combinational Systems Realization Using HDL and FPGA
	Verilog description of gates, User-defined primitives, Instantiation, Design of arithmetic functions – adder, subtractor, multiplier, divider, Implementation of multiplexer, demultiplexer, encoder and decoder, code converters, Design of complex systems, Parameterized module, Realization of test benches, Timing and delay models.
4.	Description and Design of Sequential Circuits
	Defining procedural blocks, Procedural flow control, Blocking and non-blocking assignments, Realization of latches, flip-flops, register, counter, Implementation of synchronous and asynchronous designs.

5.	Data Subsystems
	Storage modules, Memory, Stack, Queue, Functional modules, Data paths, Control subsystems.
6.	FSM and I/O Subsystem
	Finite state machine, Mealy and Moore designs, System Task and functions, Operations related to I/O subsystems, Compiler directives, VeSPA (Very Small Processor Architecture) processor—developing behavioral and structural modeling using HDL and FPGA, Design of practical systems.

Books:

1. Samir Palnitkar, Verilog HDL: A Guide to Digital Design and Synthesis, Prentice Hall.
2. Stephen Brown and Zvonko Vrsanec, Fundamentals of Digital Logic with Verilog Design, McGraw Hill.
3. Zainalabedin Navabi, Verilog Digital System Design, McGraw Hill.
4. Vivek Sagdeo, The Complete Verilog Book, Kluwer Academic Pub.
5. Peter R. Wilson, Design Recipes for FPGAs, Elsevier.

Assessment Method: Quizzes, 1 Project, In-Semester Examinations and a Final Examination will be conducted during the course.

Grading Policy:

In-Sem 1: 20%

In-Sem 2: 20%

End-Sem: 30%

Lab: 10%

Project, Quiz: 20%

Course Outcomes:

After successful completion of this course,

1. The students shall be able to define specifications and then design digital system conforming to those specifications. They shall be able to implement these designs efficiently using Verilog HDL.
2. Extensive hands on training on FPGA based projects shall give confidence to the students to realize digital circuits on the hardware. This shall facilitate students to make state-of-the-art practical research projects and meet industrial demands.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X	X							X